

Memory Map

Overview

MemoryMap is a mobile, location-based, memory sharing app that allows users to post and revisit meaningful memories tied to physical locations. The idea behind the app was to expand the concept of saving a location on an interactive map by letting users add “posts” with images, titles, and captions so that each saved location captures the user's personal experience. We wanted users to be able to both remember their own meaningful memories and see those that their friends have posted as well. The goal of this app was to connect memorable photos to the places and experiences behind them, rather than letting them get buried in a camera roll. MemoryMap addresses this by linking memories to geographic locations, which allows users to relive experiences in a more interactive, connective, and meaningful way. Users can drop memory pins at their current location, view past memories on a map, and browse their memory list. They can also interact socially through adding friends on the app and interacting with their memories.

Minimal Goals

Map Screen

The map screen displays the user's current location in real time. We wanted users to be able to drop pins directly on the map where they create memories and allow each pin to store the memory they posted. The goal with this feature was for users to be able to search locations on this screen and interact with the map. The pins would be saved in the backend database and remain there on a user's account.

View Memory Posts

This feature's goal was to allow users to both create memories and view them later. We wanted users to be able to create memories by uploading photos, notes, and a description associated with a location. We wanted these memories to be saved in a database and stay associated with the user's account. This ensures that memories can be retrieved later and associated with the correct user accounts. We also wanted these memories to be able to be viewed later by the user.

Location Detection

The location detection features let the app identify when a user is physically near a previously saved memory's location. We used background location tracking and proximity calculations based on the user's geographic coordinates. The goal with this feature was to continuously

monitor the user's location and compare it to previously saved memory coordinates. When the user is within a defined radius around a memory location, the system sees the proximity and prepares a trigger for other features.

Log In / Registration

The log in and registration page allows users to either sign up for a new account or log into an existing account once they enter the app. Our goal with this authentication system was to ensure that all memories are associated with a specific user and users' data stays private to themselves. This would allow us to create personalized memory storage and set up the app for future social features like friends and shared posts.

Push Notifications

The push notification system alerts users when they are near a location where a memory has been created. The feature relies on the location detection system and background location detection services to determine when a user enters a memory proximity zone. The goal was to send a notification when a user is near a memory location without spamming repeated alerts for the same location over and over again. The notifications are only triggered once per location event.

Stretch Goals

Friends/Private Memories

The friends and private memories system allows users to add friends and share memories with them directly through the app. The feature relies on user authentication to link accounts together so that friend connections are tied to a specific user's profile. When dropping a pin the user can select whether the memory is private or visible to friends. If friends is selected, only friends of the user would be able to view and interact with that memory pin. The goal was to give users control over who sees their memories while still encouraging sharing within a trusted social circle.

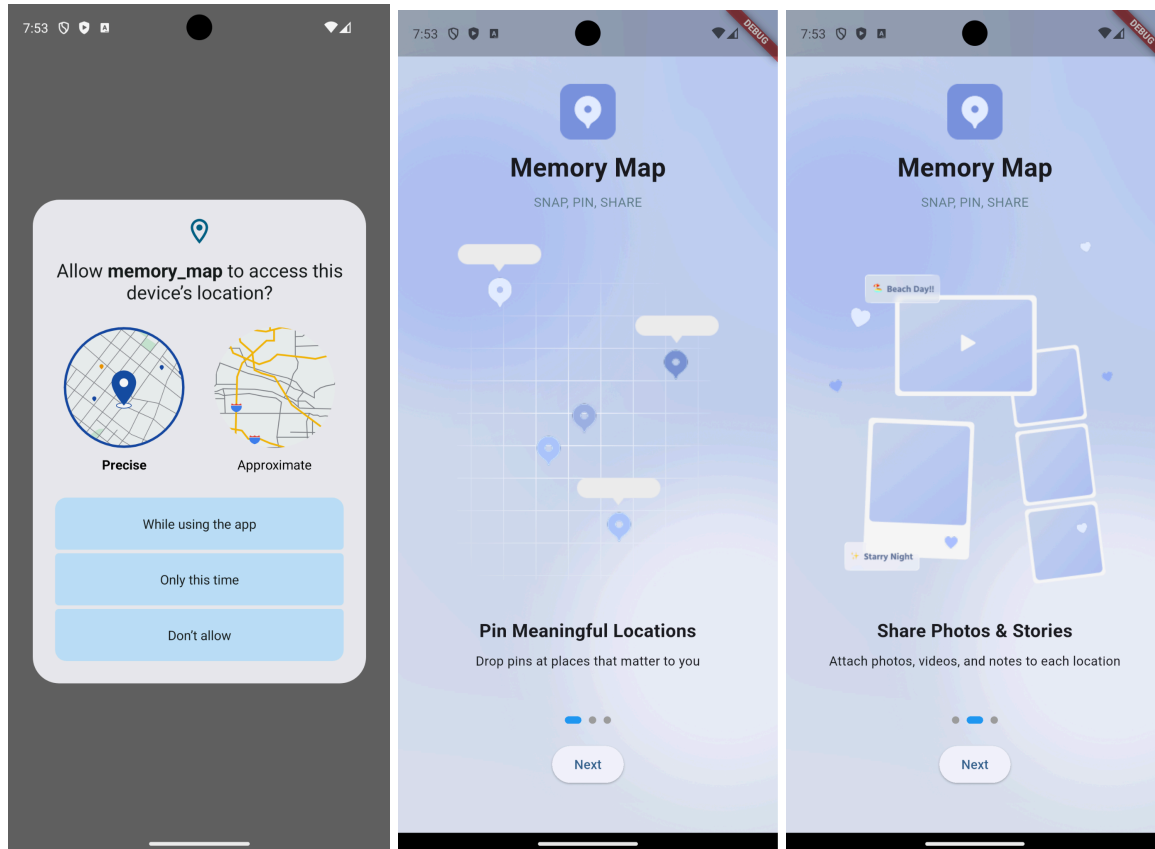
Save Memories

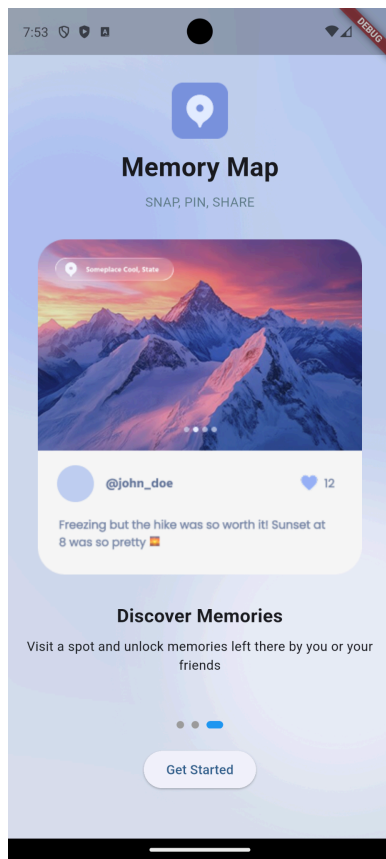
The save memories system allows users to favorite memories so that they can be easily accessed later without scrolling through their timeline. The feature relies on the existing memory storage system to flag specific pins as favorites and show them in a dedicated section of the app. When viewing a memory pin, the user can press the favorite button to save it to their collection. If the user would like to remove it, they can press the favorite button again to unfavorite. The goal was to allow users to curate their most meaningful memories and highlight memories posted by their friends.

User Interactions

Description of Onboarding Page:

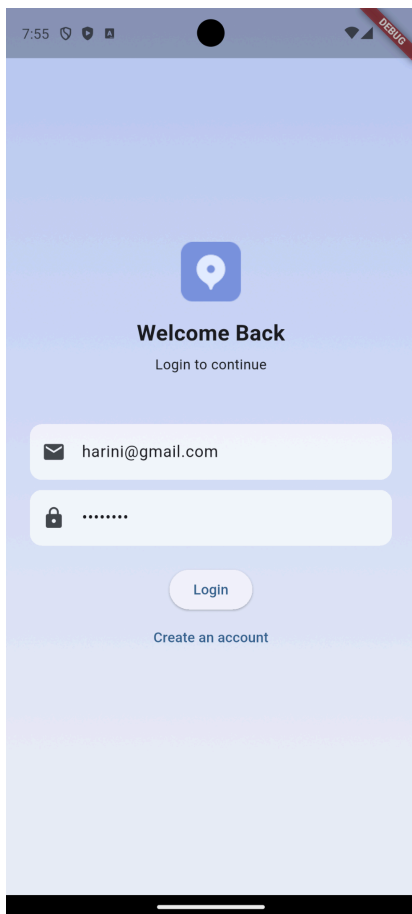
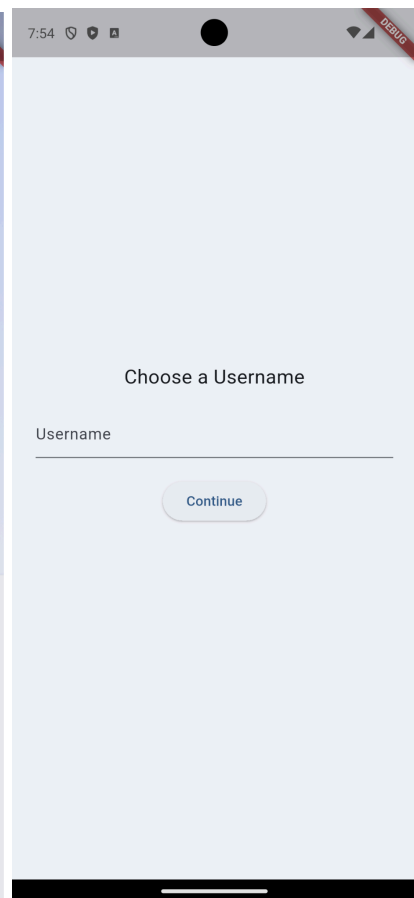
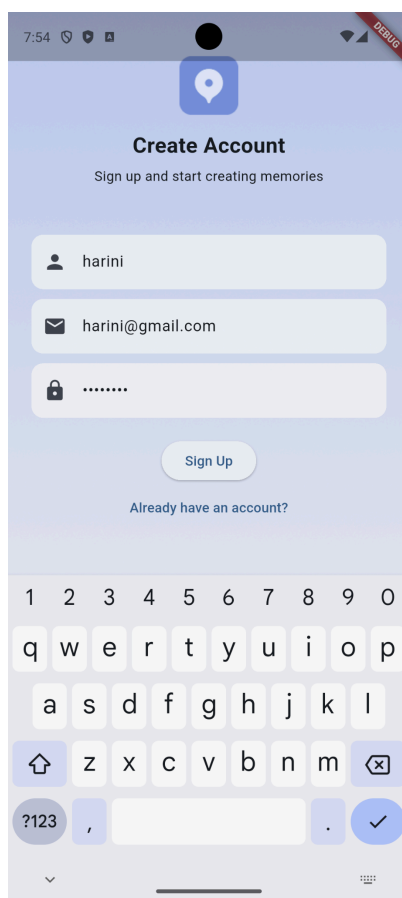
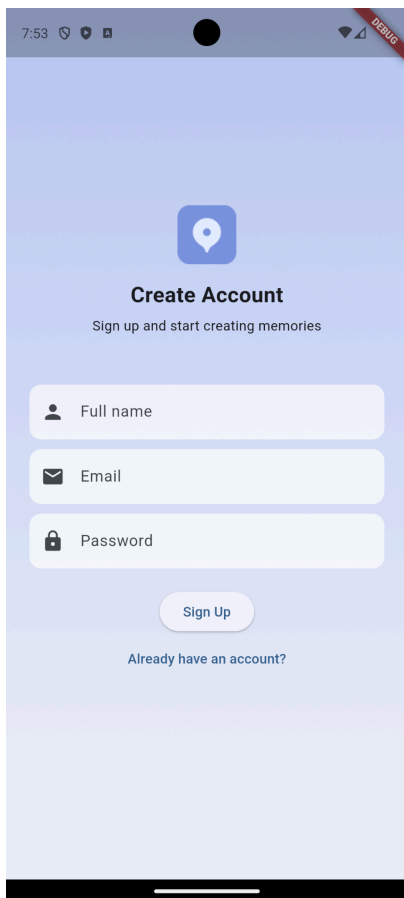
The onboarding page introduces users to MemoryMap and helps them enter the app's login and authentication flow. At first, it always asks users for permission to use their location. It's the entry point into the app where users are prompted to log in with their email before accessing any features. The second screen is a carousel of images describing what the app does. It first begins with an image describing how the user can pin meaningful locations. The second describes how they can share photos and stories. And the third shows how they can discover others' memories. After the user clicks next through all pages, it takes the user to the log-in/registration page.





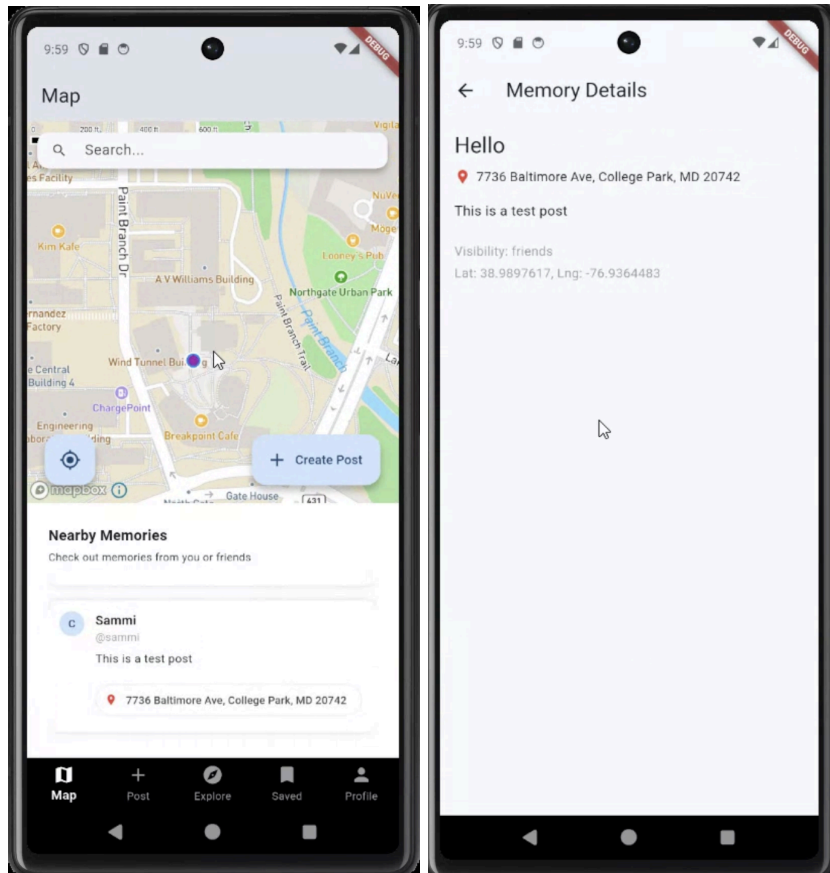
Description of Log-In/Registration Page:

The log-in/registration page allows users to either sign into an existing account or create a new one. The log in starts on the registration page where users can input their information to create an account. Users also have the option of navigating to the log in page if they already have an account and can switch back if necessary. After they are successfully either signed up or logged in, they are taken to the main app. The first screen below is where the user is taken to create an account. Here they may enter their email and password with their name to register and sign up or they can click that they already have an account. Then they are taken to choose a user name if they are registering. At this screen users are also given the option to log in to an existing account. Here they can enter their email and password and be logged in.



Description of Map Screen Page:

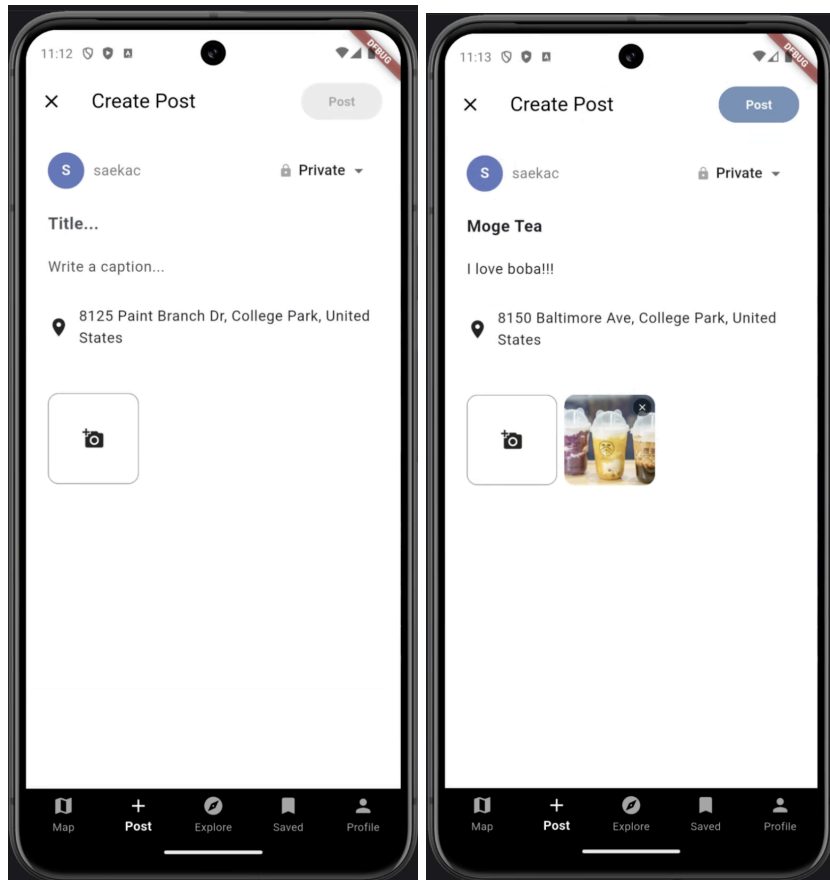
The map screen page starts with a map that users can interact with. There will be a pin at the user's current location and pins at any location where the user has previously saved posts (if there are any). There is also a search bar at the top of the map screen where users can search for any location and the map will take them to that location. The bottom of the screen will have a nearby memories list full of memories nearby to the user's current location. When the user clicks on any of these memories, it will take them to a more detailed view of the post.



Description of Make Post Page:

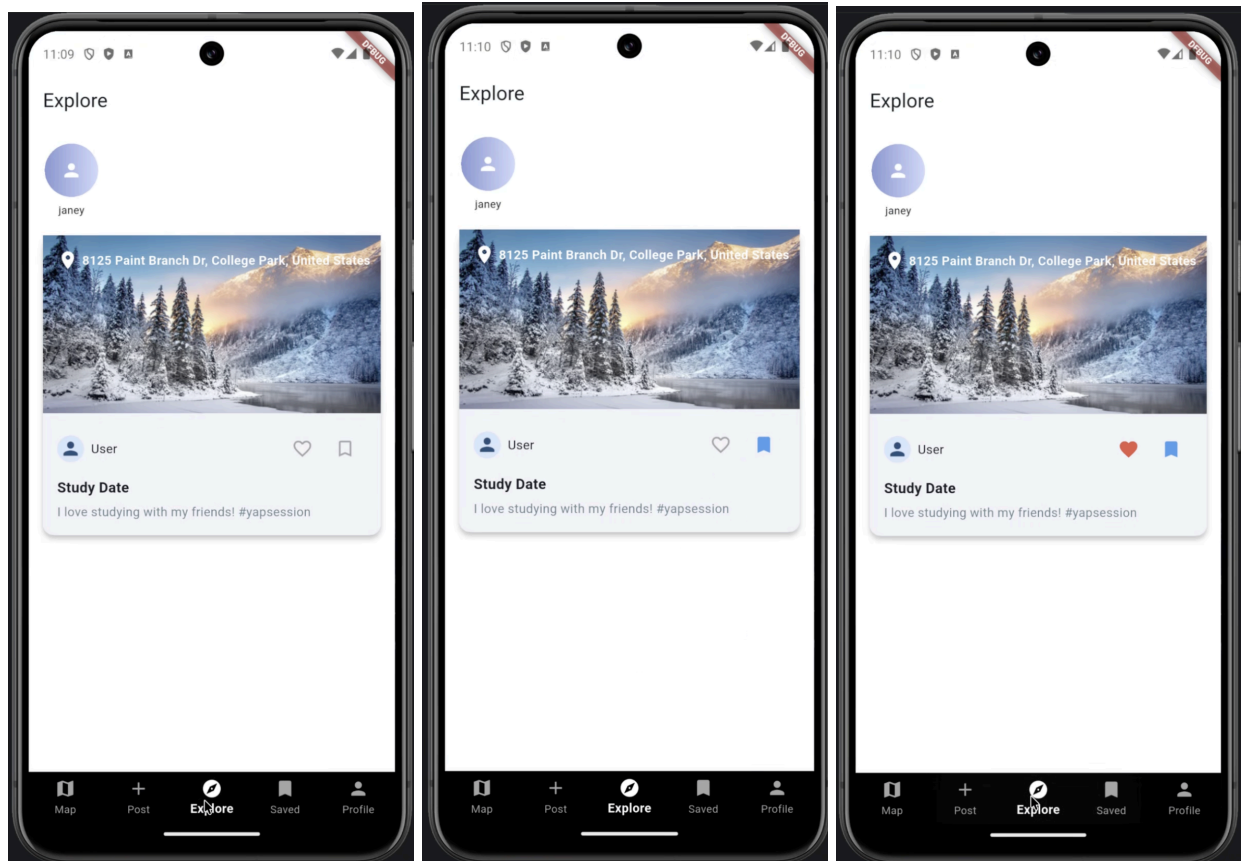
The make a post page takes users through filling out all the features of a post. The first screen has all fields blank aside from the location that is automatically detected (first image). Users are able to create and upload a new memory by entering details, attaching images, and automatically linking the memory to their current location. At the top of the screen the user sees their profile indicator, including a profile circle with their initial. To the right they have a menu that lets the user select the visibility of the post and choose between whether the post will be visible to friends or private. Below this, the user is prompted to enter a title for the memory then they are prompted to enter a caption or description for the memory. Next the user sees a location section and the app will automatically detect the user's current location and fill in the location field with the address. The user can also edit this manually if needed. Further down, the user can upload an image with the add photo icon. All of these features are seen used in the second image below. After they are selected, each image has the option of being removed with the small remove

button. Then the user can click the post button and the memory will be posted.



Description of Explore Page:

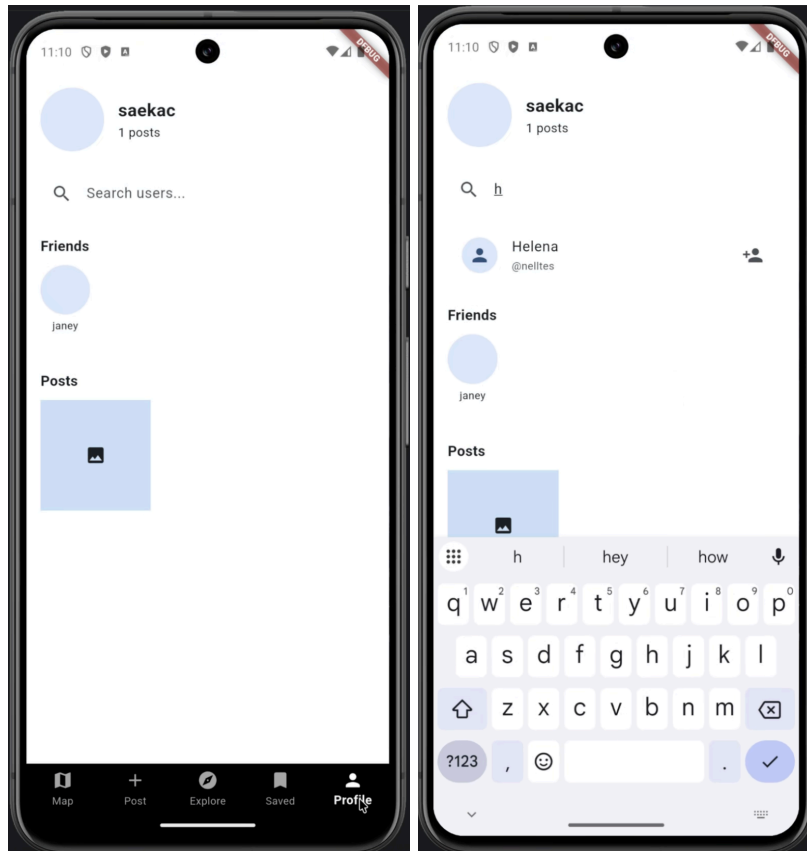
The explore page allows users to browse and interact with memory posts created by other users. It acts as a social feed within the app and displays a collection of memories that the user has permission to view based on privacy settings and friendships. When a user opens the page, they see a list of friends at the top and a list of memory posts underneath. Each post is displayed as a card containing key information such as images, title, description, and location. The system ensures that public posts are visible and friends-only posts are shown only if the user is connected as a friend to the creator. Users can interact with each post through engagement features like the like button and the save button. If there are no posts to display, the explore page shows a message that no memories are currently available. The explore page shows whether the post is saved, favorited, or both (all three images below).



Description of Profile Page

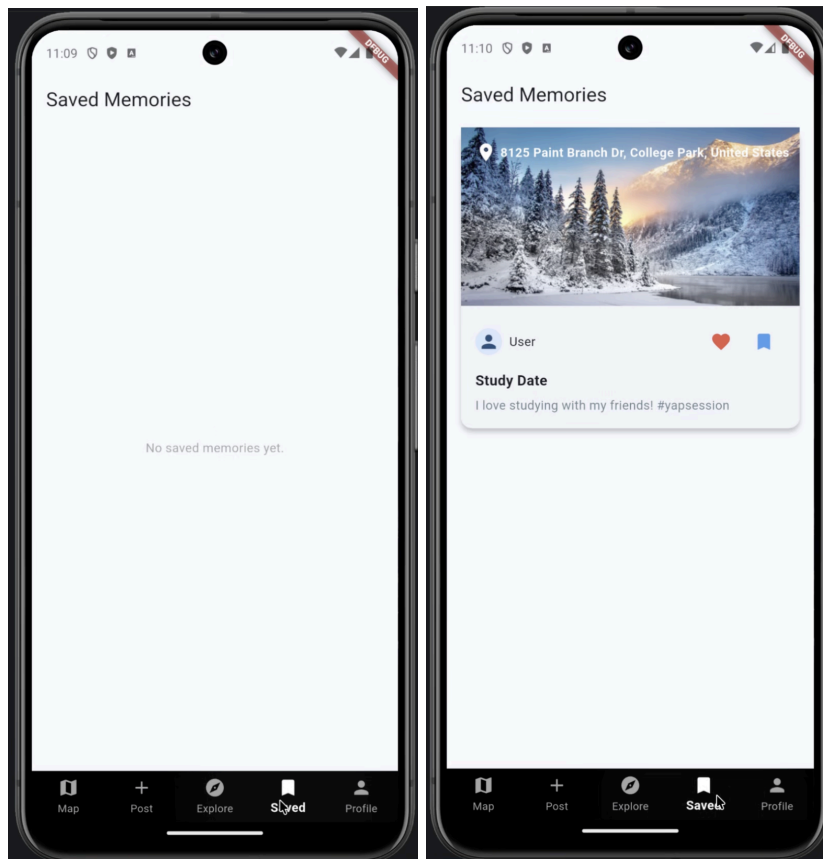
The profile page is a user-specific page that displays their personal account information. They can view their identity within the app (including their name, their own created memories, and manage their social connections). The profile page displays memories and after they upload a new memory, it appears with the associated image, note, and location. This allows users to immediately see and confirm their uploaded content and revisit past memories. The profile page also includes a friend management feature that allows users to add other users by user ID. Once a valid user ID is entered, a friend addition is processed and enables social connectivity within the app. This feature allows shared memory interactions such as viewing friends' posts. The second

image shows the ability to search for friends and add them through the profile page.



Description of Saved Memories Page:

The saved memories page displays all of the memories the user has bookmarked. When the user navigates to the page, their saved memories are displayed as a scrollable list of memory cards. Each card shows the relevant details of the memory. If the user has not saved any memories yet, they will see a bookmark icon with the message “No saved memories yet” (first image). If the user has saved memories they can scroll through their list to revisit any memory they have bookmarked (second image).



Development Process

Milestone 1

During Milestone 1, we focused on building the foundational structure of the app and implementing the core features required for basic functioning before building on top of that. This included things like setting up the map interface and screen, creating a template for memories to be created, and setting up a database for storing many necessary features (like accounts, friendships, posts).

One of the main tasks in this phase was creating a map screen and view into the app. We originally created a static version of the map tab just to establish the layout and navigation structure and planned on expanding the actual functionality later in the project. Another main part of this milestone was creating the memory storage system. We originally planned on using Firebase but this didn't align well with the data we wanted to store and it wasn't as accessible to us. We then decided to set up our database system using Supabase to handle user authentication, post memory data, etc. Another feature we worked on was the memories list tab. We set up a basic structure for this so we could expand on it later. This included structuring the input flow for adding a photo, note, and location, and defining a consistent way to pass this data to the backend.

Some setbacks during this milestone were early setup issues like the emulator being unstable with our project and coordinating how data would flow between the frontend and backend. We

also had some difficulty around database schema design since it required us to ensure compatibility with future features.

Milestone 2

During Milestone 2, we focused on implementing the notification handling system, integrating location-based interactions into the app, and expanding on several other pages in the app. This phase was mostly centered around determining when users should receive notifications, how those notifications are triggered, and how to ensure they enhance the user experience without becoming overwhelming and spamming the user at a certain location. So, a significant part of this milestone was developing the logic for those features.

Despite this progress, we still had a lot of integration challenges. At this stage, the full notification system was not yet completely connected to the backend, which made it difficult to test notifications. We also had trouble placing pins on the map for saved memories and inconsistencies in displaying saved memories within the memories tab. As we tried to expand on the camera feature for the memory posts, we also ran into trouble connecting that with the Supabase. These problems were mostly due to incomplete integration with Supabase and mismatches between frontend expectations and backend data structures. While we had some minor setup issues earlier in development, they were resolved during this phase. Based on these challenges, several adjustments were made to the original plan. Instead of immediately finalizing all notification features, we prioritized building a reliable foundation for notification handling and spam prevention, with the understanding that full integration would be completed in the next milestone.

Milestone 3

During Milestone 3, we focused on fixing existing bugs and implementing several new features across the app. On the fixes side, we resolved issues with loading and adding pins on the map, corrected database errors that were preventing saved memories from displaying correctly, and reduced our memory visibility schemas down to private and friends only.

For new features, we implemented the logic for user login and authentication, including storing the user ID locally on the device and in a user table in our database. We also worked on camera integration, allowing users to upload images from their gallery or take a photo directly in the app, which were then uploaded to our backend storage and pulled from our posts table. On the notifications side, we finished implementing push notification logic, including sending users to the login screen if they were not already signed in, or directly to the relevant memory screen if they were. We also built out the explore page, which allows users to see posts that their friends have recently made and heart any posts they like.

For the saved memories feature, we created a table to store saved memories where each entry tracks the user who saved it and the post that was saved. We also built the profile page, which displays the user's profile information, friends list, and their own posts on one scrollable page.

Users can search for and add friends by username, view their current friends list, and see their own posts displayed as cards showing the title, description, and location of each memory.

We ran into several setbacks throughout this milestone. Emulator issues made it difficult to consistently test changes throughout the development process. We also had a lot of challenges with image uploads and figuring out how to properly store and retrieve media files through our backend. Push notifications were difficult to get working reliably, and we had numerous merge conflicts when combining everyone's work which caused additional delays. On the database side, we ran into errors stemming from table relationships that were not properly configured, which caused queries to fail when trying to retrieve saved memory data. The explore page also presented issues as it was not displaying posts as expected, which required additional debugging before it could function correctly.

Future Directions

While MemoryMap covers the core experience of saving and revisiting location-based memories, there are several features that could further the experience of the app that we would implement with more time. One direction would be to introduce shared memory groups, where users could contribute memories to a collective map that all group members could view and interact with (essentially like a shared photo album). This would shift the app from a personal journaling experience to a collaborative one, allowing friends or families to build a shared archive of places they have been together. Another feature would be to introduce a travel statistics page that gives users a summary of their activity on the app, such as a total distance traveled between memory locations, number of memories created over time, and a heatmap of their most frequently visited areas. Additionally, it would be valuable to implement a search and filter system within the memories list so that users can search by date, location, or keyword rather than scrolling through their entire history. Finally, we would want to expand the notification system to allow users to customize their proximity alert radius and notification preferences per memory would give users a greater sense of control over their overall experience.